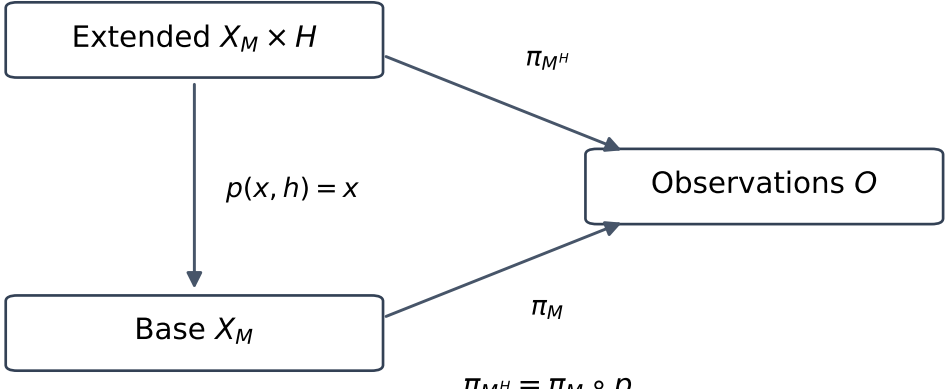
Extended $X_M \times H$
π_{M^H}
$p(x, h) = x$
Observations O
Base X_M
π_M
$\pi_{M^H} = \pi_M \circ p$